Working paper | Peer reviewed · report published | 13 October 2026

Four micropeptide-source RNAs fail as a colon cancer recurrence panel, and the marker that survives clears a selection-aware null only narrowly

A panel picked for mechanism rather than by screening, tested against a thousand random panels put through the same machinery — on eighteen recurrences. The one marker that beat it now has a null of its own, and passes by four per cent.

Leo Lee[1]

1 Shanghai High School International Division, Shanghai, China | *Working paper*

PRISM-WP-003 (2026)

137 patients, 18 events **2,000** random comparators **8** figures PRISM-WP-003

ABOUT THIS WORKING PAPER

Peer reviewed by Henry Wang, who is not independent of the author. The signed report is published in full: minor revision, with scores of 4, 5, 4 and 5 for sound, honest, checkable and legible. The referee is PRISM's editor-in-chief, is at the author's school, and is the author's co-author on PRISM-1-004. PRISM draws its reviewers from one school whose authors routinely write together, so independent refereeing is frequently not available to it; the journal's rule is disclosure rather than avoidance. Read the report as informed rather than independent.

The paper remains a **working paper** rather than a numbered article because no handling editor has recorded a decision, and the only available editor is the referee. The revisions the report asked for were made — by the journal, not by the author.

The framing of this version is not the author's, and one analysis is not his either. The submitted manuscript carried an AI-assisted abstract, discussion and conclusion and a

declaration that the author would rewrite them before submission; that rewrite has not happened. The title, abstract and the prose of sections 5 to 13 were written by Claude (Anthropic) for the journal. None of the author's results were altered and no model of his was refitted.

The submitted study identified a missing comparison — a random single-gene null for LINC00675 — as the gap that left its headline number uninterpretable. The journal computed it (§4.1, §10, Fig. 8), after requiring the re-implementation to reproduce all four of the author's published single-marker C-indices and all 100 of his per-fold values exactly. The proportional-hazards and stage-adjustment analyses in §11 were added the same way. Those computations are the journal's, not the author's; their code is deposited separately so it can be checked or discarded on its own.

A referee report exists, and is deliberately not published as a PRISM review. Henry Wang, editor-in-chief, filed a signed report on the working-paper version recommending minor revision, and this version answers it. He is barred from refereeing this manuscript under PRISM's conflict policy — he is at the author's school and is the author's co-author on PRISM-1-004 — so the report is not published as a peer review, and the paper stays a working paper until an eligible referee reports or the policy is changed. The revisions stand on their own merits either way.

Abstract

Transcripts once classified as long noncoding RNAs can encode functional peptides, and several of those peptides have documented roles in colorectal cancer. This study asked whether the source RNAs of four experimentally characterised micropeptides — LINC00467 (ASAP), ELFN1-AS1 (SMIMP), LINC00675/TMEM238L (FORCP) and HOXB-AS3 — form a useful recurrence panel in colon adenocarcinoma.

They do not. In 137 stage II/III TCGA-COAD patients with 18 disease-free interval (DFI) events, the four-transcript elastic-net Cox panel reached a held-out C-index of 0.609 (95% CI, 0.509–0.717) under five-by-three nested cross-validation repeated over five seeds, below the 0.65 development target set in advance. Against 1,000 random four-gene panels put through the identical procedure, it ranked at the 93.4th percentile, one-sided empirical $p = 0.067$. The prespecified comparison fails at its own threshold.

The individual transcripts are the more useful result. Three of the four discriminated at or below chance for primary DFI: 0.462 for LINC00467, 0.367 for ELFN1-AS1 and 0.515 for HOXB-AS3. Across the three cohorts and four transcripts, 7 of 12 estimates fall below 0.5.

Only LINC00675 stood out, at 0.700, and it was the sole transcript retained in the final penalised refit. Its interval of 0.583–0.819 is a patient-cluster bootstrap *conditional on the fitted cross-validation models*: it does not repeat model fitting, hyperparameter selection or the choice of this marker, so it is narrower than the uncertainty a reader should attach to the number.

The study as submitted had no way to judge that 0.700, because it built a null for the panel and none for a single gene. We computed the missing null: 1,000 genes drawn at random and put through the same folds, the same measurement gate and the same unpenalised Cox model. Restricted to the 279 draws that pass the gate in all 25 folds, as all four candidates do, LINC00675 sits at the 98.9th percentile ($p = 0.014$), just above the null's 99th percentile of 0.699. Correcting for its having been the best of four candidates, a result this good arises by chance **4.2%** of the time. The marker therefore clears its first real test, and clears it narrowly: a selection-aware empirical calibration of 0.042, not a four-point margin over any biological benchmark. Proportional hazards is not contradicted for any of the four transcripts, and the LINC00675 coefficient is unchanged by stage adjustment — though stage itself predicts nothing in this cohort, so that last check proves less than it sounds.

Two features of the comparator distributions bear on both tests. Of the 1,000 random panels, 264 (26.4%) returned a C-index of exactly 0.500 because the penalty shrank every coefficient to zero, leaving constant risk; among the 736 that fitted anything, 9.0% matched or beat the candidate panel, a one-sided p of 0.091 rather than 0.067. The same effect is far stronger for single genes, where 710 of 1,000 draws produce no model at all, which is why the unmatched single-gene null ($p = 0.004$) overstates the case and the expression-matched one is the figure to quote.

What remains true is that the panel strategy failed, that most of these mechanistically chosen transcripts carry nothing, and that the one that does still rests on 18 events, shows nothing for overall survival, and has never been tested outside this cohort or against clinical risk factors.

Keywords: colorectal adenocarcinoma; LINC00675; TMEM238L; FORCP; micropeptide-source RNA; disease-free interval; TCGA; negative result.

1. Micropeptide-source RNAs as candidate markers

Small open reading frames in transcripts historically classified as long noncoding RNAs can encode functional peptides. Ribosome-profiling atlases have expanded the catalogue of human translated regions, and proteomic studies have supplied direct evidence for subsets of noncanonical translation products [1,2,3]. These approaches distinguish transcription, translation, peptide detection and biological function. Expression of a source RNA is therefore a tractable candidate biomarker, but it cannot by itself establish the abundance or the activity of the encoded peptide. Everything below concerns RNA.

Several experimentally characterised peptides motivate a colorectal cancer panel. LINC00467-encoded ASAP interacts with ATP synthase and promotes mitochondrial respiration and tumour growth [4]. ELFN1-AS1-encoded SMIMP, identified by functional interrogation of noncanonical open reading frames, interacts with SMC1A and supports repression of cell-cycle inhibitors [5]. LINC00675-encoded FORCP localises to the endoplasmic reticulum, suppresses proliferation and tumorigenicity, and supports apoptosis under endoplasmic-reticulum stress in well-differentiated colorectal models [6]. The HOXB-AS3 peptide antagonises hnRNP A1-dependent pyruvate kinase splicing and metabolic reprogramming [7]. LINC00266-1-encoded RBRP enhances IGF2BP1 recognition of m^{6}A-modified transcripts [8].

Mechanistic plausibility is not prognostic performance. Established colon-cancer expression classifications and commercial recurrence assays were both developed against independent cohorts and assessed alongside conventional clinical factors [9,10], a standard no single-cohort analysis can meet. This study tested whether a panel chosen on mechanism rather than on a survival screen would discriminate recurrence in TCGA colon adenocarcinoma, and then asked which individual transcripts carried whatever discrimination appeared.

2. What was fixed in advance, and what was not

The distinction matters more here than the point estimates, so it is stated before the results.

Fixed before the outcome data were examined: the four candidate transcripts and their Ensembl identifiers; the primary endpoint (DFI) and primary cohort (stage II/III); the

0.65 C-index development target; the nested cross-validation design, its grids and its seeds; the 1,000-panel random comparator and its one-sided empirical probability; and the fold-support rules for time-dependent AUC.

Decided after results were seen, and therefore exploratory: the focus on LINC00675; every single-marker contrast; the paired panel-minus-marker difference; and the expression median split. These are labelled post hoc wherever they appear and are not corrected for multiplicity. Readers should treat the prespecified panel test as the result of this study and everything concerning LINC00675 as hypothesis generation.

Added after submission, by the journal rather than the author: the random single-gene null of §10. It was computed because the submitted manuscript identified its absence as the decisive gap, and it is reported with an explicit correction for the post hoc selection it is calibrating. It does not convert the exploratory finding into a prespecified one.

3. Data and cohorts

Bulk expression came from the UCSC Xena TCGA-COAD STAR-Counts matrix [11]. That matrix contains $\log_2(\text{count}+1)$ values, as its dataset metadata specifies; the filename must not be read as indicating untransformed counts. Counts were reconstructed as 2^x-1, retaining fractional values arising from upstream averaging across portions and aliquots. Clinical outcomes and staging came from Supplemental Table S1 of the TCGA Clinical Data Resource, restricted to COAD [12]. DFI and overall survival used the supplied event indicators and time fields with no outcome relabelling. Times were held in days; displayed months use 365.25/12 days.

Four source RNAs were matched to Ensembl identifiers after version-suffix removal. LINC00266-1/RBRP was considered mechanistically but excluded before any model was fitted, because no stable Ensembl identifier could be resolved in this annotation; the exclusion was logged as missing annotation, never as zero expression. Gene-level abundance does not resolve peptide-encoding isoforms.

Source RNA	Ensembl gene ID	Peptide	Documented biology (from the original studies, not from tl
LINC00467	ENSG00000153363	ASAP	ATP synthase activation, mitochondrial respiration, tumour g
ELFN1-AS1	ENSG00000236081	SMIMP	SMC1A interaction, repression of CDKN1A/CDKN2B [5]

Source RNA	Ensembl gene ID	Peptide	Documented biology (from the original studies, not from tl
LINC00675 / TMEM238L	ENSG00000263429	FORCP	ER localisation, stress-associated apoptosis, suppressed pr
HOXB-AS3	ENSG00000233101	HOXB-AS3 peptide	Antagonism of hnRNP A1-mediated PKM splicing [7]
LINC00266-1	Not resolved; excluded	RBRP	IGF2BP1-mediated m⁶A recognition [8]

Expression columns were restricted to primary-tumour specimens. The first 12 characters of each barcode defined the patient identifier. Where several primary specimens existed, the lexicographically first barcode was kept, giving one specimen per patient without consulting outcomes. The expression matrix held 514 specimens, comprising 471 primary tumours from 458 unique patients, of whom 456 matched a COAD clinical record. Stage II/III eligibility identified 305 patients; 168 lacked DFI status, leaving 137 patients with 18 DFI events. Reverse Kaplan–Meier median follow-up was 31.05 months.

Cohort	*n*	Events	Follow-up (mo)	Panel C-index	Random-panel percentile	One-sided
Primary DFI, stage II/III	137	18	31.05	0.609 (0.509–0.717)	93.4	0.067
Expanded DFI, stages I–IV eligible	181	21	30.98	0.593 (0.478–0.703)	86.7	0.134
Secondary OS, stages I–IV	428	93	27.01	0.548 (0.487–0.606)	89.9	0.102

Although expanded DFI eligibility admitted stage IV, no stage IV patient had usable DFI, so that cohort is stages I–III. Differences between endpoints cannot be separated from differences in cohort composition.

4. Statistical analysis

Counts per million used the library sum over the full expression matrix, followed by $\log_2(\text{CPM}+1)$. This transformation is sample-local and pools no information across patients. A transcript was measurable if CPM reached 1 in at least 20% of training tumours. Each training partition separately determined median imputation, removal of constant features, and mean/standard-deviation scaling. Missing annotation was never replaced with zero. No batch correction was applied across folds.

The panel used an elastic-net penalised Cox model over a penaliser grid of {0.001, 0.01, 0.1, 1, 10} and an L1-ratio grid of {0.1, 0.5, 0.9, 1.0}. Nested cross-validation used five outer and three inner folds stratified by event status, repeated with seeds 11, 23, 37, 53 and 71. Inner-fold comparable-pair-weighted concordance selected hyperparameters, with ties favouring stronger penalisation and then a larger L1 ratio. Standardised coefficients below 10^{-4} in absolute value were treated as zero and the baseline hazard recomputed. A separate full-cohort refit summarised retained coefficients and was never used to score held-out discrimination.

Harrell's C-index compared risks only within each held-out outer fold. Comparable-pair counts were aggregated across folds within a seed, and the five seed-specific estimates averaged. Cumulative/dynamic AUC used inverse-probability-of-censoring weighting with the censoring distribution estimated on the corresponding training set. A horizon was reported only if, in every outer fold, there were at least three held-out events by the horizon, five held-out observations beyond it, ten training observations beyond it, and training censoring survival of at least 0.05. A horizon failing these rules was reported as not estimable rather than averaged over the folds that happened to pass — a conservative reporting rule, not a claim that every alternative estimator is undefined.

For the random comparator, 1,000 panels of four distinct genes were drawn uniformly from the full annotation universe after excluding the four candidates, and each was put through the identical nested cross-validation, measurement gate, preprocessing and hyperparameter search. The same draws were reused across cohorts. Genes were not prefiltered on whole-cohort expression nor matched to the candidates by expression level. Percentiles used midranks; the one-sided empirical probability was $(1 + \#\{C_{\text{random}} \geq C_{\text{panel}}\})/1001$.

Each transcript was also fitted alone in an unpenalised continuous-expression Cox model on exactly the panel's outer partitions, with gate, imputation and scaling fitted on the corresponding training set. Reusing the folds permits paired comparison but does not correct the optimism of choosing the strongest marker after inspecting results.

4.1 The random single-gene null, added after submission

The null of §10 was not part of the submitted study. It draws 1,000 genes uniformly from the 60,656 remaining after the four candidates are removed, and puts each through the author's own frozen fold assignments, measurement gate, training-only imputation and scaling, unpenalised univariate Cox fit and concordance rule. A coefficient below 10^{-4} in absolute value is set to zero, which is the author's own retention convention and the reason a model that fits nothing scores exactly 0.500 rather than an arbitrary value.

Neither `lifelines` nor `scikit-survival` could be installed in the environment available, so the per-fold Cox fit and the concordance counts were re-implemented, with Efron handling of tied event times. The re-implementation was then required to reproduce the author's recorded results before it was allowed to produce a null: it returns all four published single-marker C-indices to the last decimal place recorded, together with all 100 per-fold C-indices, comparable-pair counts and coefficients across the four markers and 25 folds, and it reproduces the archived per-sample library sizes exactly. The expression matrix was re-downloaded and its SHA-256 checked against the hash recorded in the author's `results.json`, so the input is the analysed snapshot and not a later revision.

The matched null restricts the 1,000 draws to those passing the measurement gate in all 25 outer folds, which all four candidate transcripts do. That criterion uses expression only and no outcome, so it introduces no selection on the quantity being tested. Because LINC00675 was reported as the best of four candidates, the selection-aware figure is the proportion of resamples in which the maximum of four genes drawn from the matched null reaches or exceeds its C-index, over 200,000 resamples. The four genes are drawn without replacement, since the candidates are four distinct transcripts, and a resample counts whenever its maximum is greater than *or equal to* the observed value. Drawing with replacement instead changes the figure by less than the Monte Carlo error: 4.21% without, 4.19% with, against a Monte Carlo standard error of about 0.05 percentage points.

This calibration treats the four mechanistically chosen candidates as four draws from the matched random-gene population. That is a defensible reference, not an exact model of how they were chosen, so the resulting 4.2% is reported here as a **selection-aware empirical calibration** and not as a multiplicity-corrected significance level. Positive correlation among the four candidates would make it conservative; any looking beyond these four that went unrecorded would make it optimistic.

Percentile 95% confidence intervals used 1,000 patient-cluster bootstrap replicates, each patient's multiplicity shared across all repeated held-out predictions. These intervals condition on the fitted models and training censoring distributions: model fitting, hyperparameter tuning and marker selection were **not** repeated inside the bootstrap, so the intervals understate total uncertainty.

5. The panel did not beat its own null

The primary panel reached a held-out C-index of 0.609 (95% CI, 0.509–0.717), below the 0.65 target fixed in advance. It ranked at the 93.4th percentile of the 1,000 random four-gene panels, one-sided empirical $p = 0.067$. The mean random-panel C-index was 0.502. All 1,000 random panels were retained in the comparison; two failed inner candidate fits in a single random panel were handled by invalidating that hyperparameter candidate rather than dropping the panel.

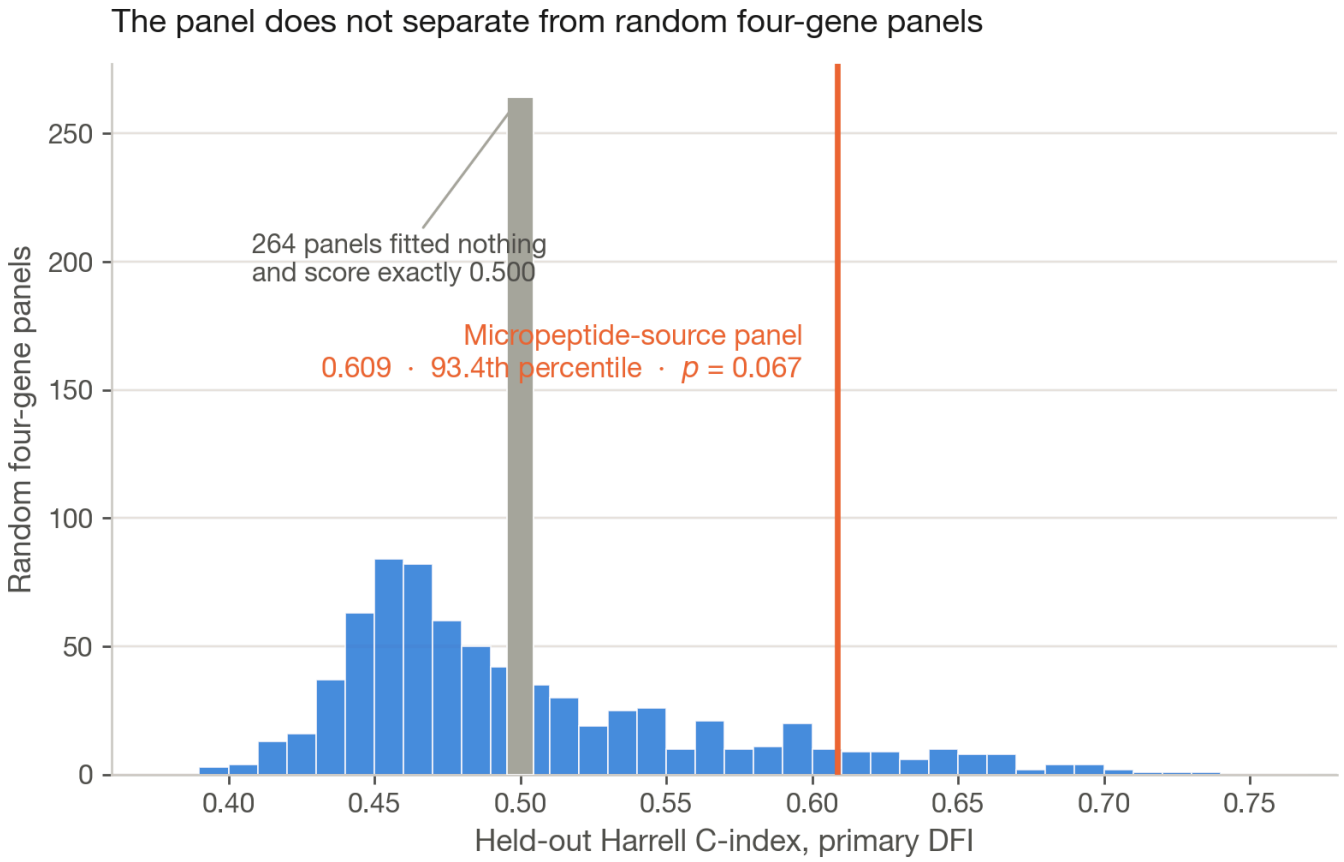


Fig. 1 | Primary-DFI C-indices for 1,000 random four-gene panels put through the identical nested cross-validation. The line marks the micropeptide-source panel at 0.609, the 93.4th percentile, one-sided empirical p = 0.067. The grey column is drawn separately because it is not part of the distribution around it: 264 panels had every coefficient shrunk to zero, leaving constant risk and a C-index of exactly 0.500 by construction. §5.1 reports the comparison restricted to the 736 panels that fitted something.

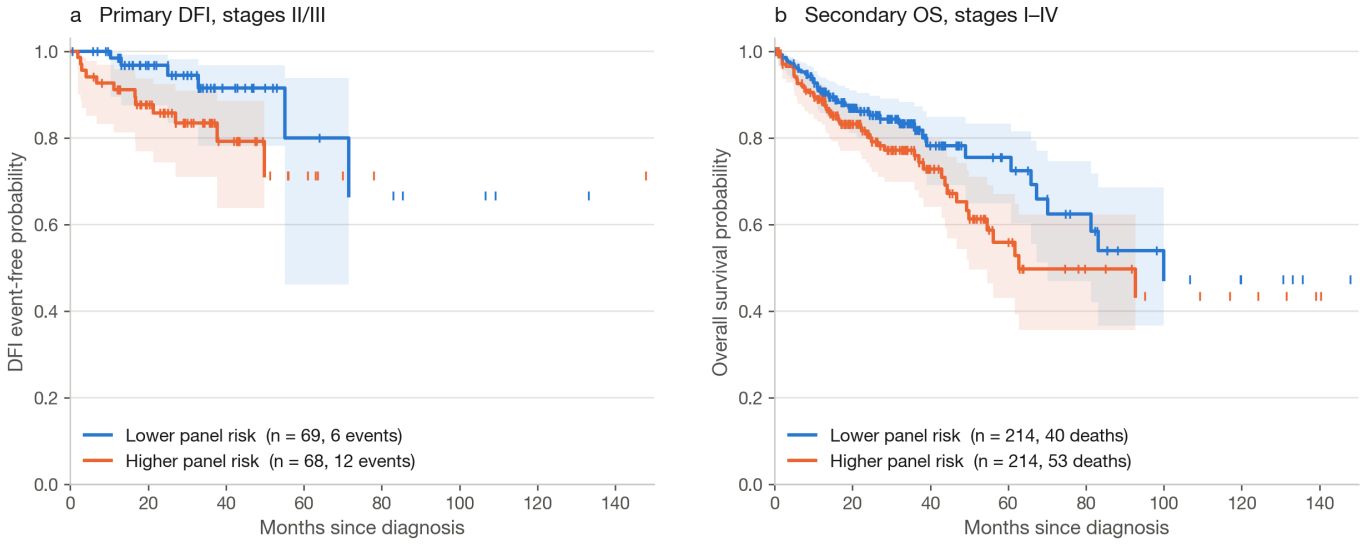


Fig. 2 | Descriptive Kaplan–Meier curves by four-transcript panel risk for (a) primary DFI, stages II/III, and (b) secondary overall survival, stages I–IV. Each held-out patient's risk percentile was computed against the corresponding outer-training distribution, averaged across repetitions and split at the cohort median. Shading is pointwise 95%; ticks mark censoring. Not a validated risk stratification.

The same conclusion holds in both secondary cohorts, with larger probabilities: 0.134 for expanded DFI and 0.102 for all-stage OS. A mechanistically chosen panel that cannot separate itself from randomly drawn genes put through the same machinery has not demonstrated that the mechanism carries prognostic information in this dataset.

5.1 A quarter of the null distribution is inert

The null in Fig. 1 has a conspicuous spike at exactly 0.500. It is not a rendering artefact. Of the 1,000 random panels, **264 (26.4%) returned a C-index of exactly 0.500**, because for those panels the selected penalty shrank every coefficient to zero, leaving a constant risk score and therefore, by definition, a concordance of one half. These panels are not weak competitors; they are models that declined to fit.

This matters for the headline comparison. The published probability treats all 1,000 draws as comparators, giving 66 at or above the panel and $p = 0.067$. Restricted to the 736 panels that fitted anything at all, 66 of 736 (9.0%) match or beat the candidate panel, a one-sided probability of 0.091. The candidate panel's rank is flattered by a comparator distribution that is a quarter inert.

Both figures exceed 0.05 and the prespecified conclusion is unchanged, so this does not rescue or reverse anything. It is reported because the percentile is the paper's headline comparison and a reader is entitled to know that roughly a quarter of the distribution it is measured against consists of null models in the strict sense. The effect works against the panel, which is why the negative result should be read as firmer rather than more fragile.

6. What the elastic net kept

Only LINC00675/TMEM238L survived the final full-cohort refit, at penaliser 0.1 and L1 ratio 0.5. That final model must be distinguished from the models that produced the held-out predictions: across the 25 outer folds, LINC00675 was retained in 23, HOXB-AS3 in 18, LINC00467 in 16 and ELFN1-AS1 in 12. All four passed the measurement gate in every fold. The panel's C-index therefore describes a collection of fold-specific penalised models, not repeated fitting of one fixed model.

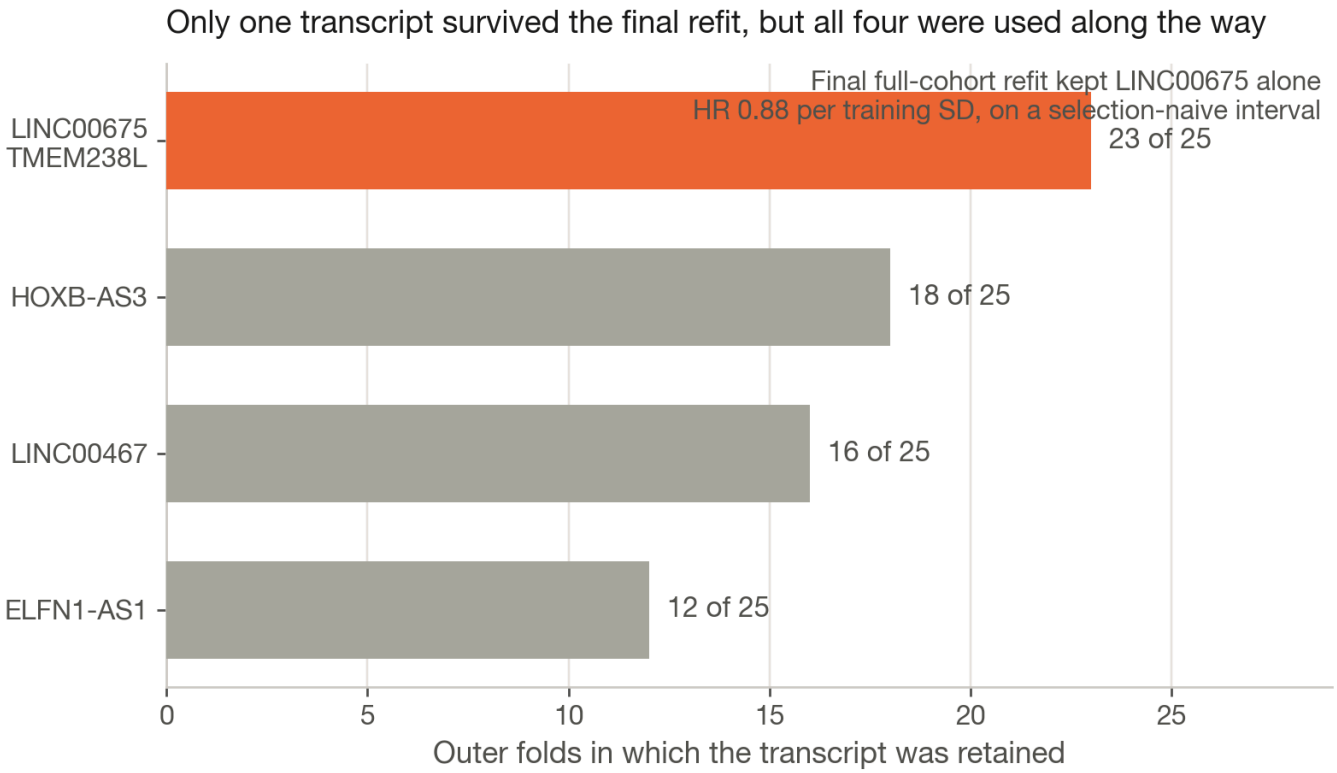


Fig. 3 | How often each transcript was retained across the 25 outer folds. The final full-cohort refit kept LINC00675/TMEM238L alone, at a shrunken hazard ratio of 0.88 per training standard deviation on an interval that ignores both variable selection and penalisation, so it is not confirmatory. The held-out estimates in this paper come from the fold-specific models counted here, not from that refit, which is why the other three transcripts matter even though none survived it.

7. Three of the four transcripts were at or below chance

The single-marker benchmarks are the most informative part of this study, and they are not encouraging for the panel concept.

Source RNA	Primary DFI (II/III)	All-stage DFI	All-stage OS
LINC00467	0.462 (0.343–0.582)	0.400 (0.314–0.488)	0.431 (0.394–0.469)
ELFN1-AS1	0.367 (0.305–0.436)	0.479 (0.353–0.596)	0.588 (0.519–0.655)
LINC00675 / TMEM238L	**0.700 (0.583–0.819)**	0.691 (0.571–0.802)	0.502 (0.435–0.563)

All intervals are patient-cluster bootstrap intervals conditional on the already-fitted cross-validation models. They do not include the variability of model fitting, hyperparameter selection, or of having chosen the marker in the highlighted row after seeing this table.

HOXB-AS3	0.515 (0.380–0.652)	0.573 (0.423–0.715)	0.451 (0.420–0.484)

For primary DFI, three of four transcripts sat at or below chance. Across all twelve transcript-by-cohort combinations, seven estimates fall below 0.5, and LINC00467 is below 0.5 in all three cohorts. Choosing transcripts because their peptides have documented colorectal biology did not, in this dataset, yield transcripts that carry prognostic signal.

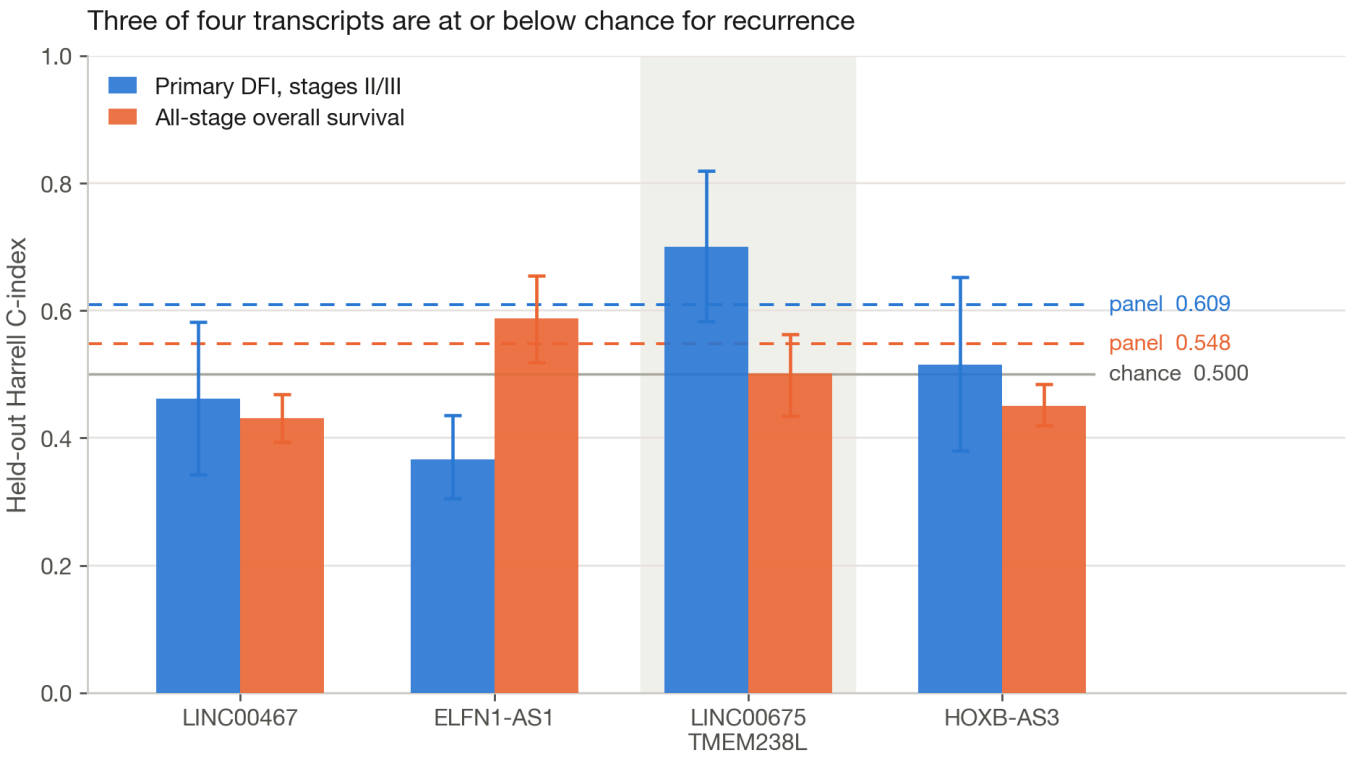


Fig. 4 | Held-out single-transcript Cox discrimination for primary DFI and all-stage overall survival, with 95% intervals from 1,000 patient-cluster bootstrap replicates of held-out predictions without refitting. Dashed lines mark the corresponding panel estimates. Three of four transcripts sit at or below chance for DFI, and the DFI and OS orderings invert.

Against that background, LINC00675's 0.700 for primary DFI is the single exception among four mechanistically motivated candidates. Its interval in the table is conditional on the fitted models, as §4 explains, and is not a full inferential interval. The paired panel-minus-LINC00675 difference was −0.092 (95% CI, −0.192 to −0.007), conditional on the fitted models: the four-transcript strategy was worse than its own best component. That contrast is post hoc, uncorrected for multiplicity, and conditional on models the bootstrap did not refit.

8. Stage, endpoint and horizon support

In the 181-patient expanded DFI cohort with 21 events, LINC00675 reached 0.691 (95% CI, 0.571–0.802) against a panel C-index of 0.593. This cohort contains the primary cohort and is a stage-sensitivity analysis, not independent replication.

The endpoint contrast is the sharpest internal check available, and LINC00675 does not survive it. In the 428-patient OS cohort with 93 deaths — more than five times the primary cohort's events — LINC00675 achieved 0.502 (95% CI, 0.435–0.563), indistinguishable from chance. ELFN1-AS1, which was the *worst* DFI marker at 0.367, became the best OS marker at 0.588. A marker ordering that inverts between endpoints in the same resource is what one expects from noise, though cohort composition, mortality and treatment also differ between the two analyses and cannot be separated here.

Cohort	Horizon (yr)	Events	Beyond horizon	Censored by horizon	AUC (95% CI)
Primary DFI	3	14	46	77	Not estimable
Primary DFI	5	17	13	107	Not estimable
All-stage DFI	3	17	55	109	Not estimable
All-stage DFI	5	20	16	145	Not estimable
All-stage OS	3	70	109	249	0.543 (0.459–0.620)
All-stage OS	5	83	45	300	Not estimable

Only three-year OS AUC met the support rules, at 0.543 (95% CI, 0.459–0.620) — an interval comfortably containing 0.5. Both DFI horizons failed support in 8 and 25 of 25 folds at three and five years; five-year OS failed in one fold, where only four held-out patients remained beyond the horizon. Having more events did not guarantee adequate support in every fold.

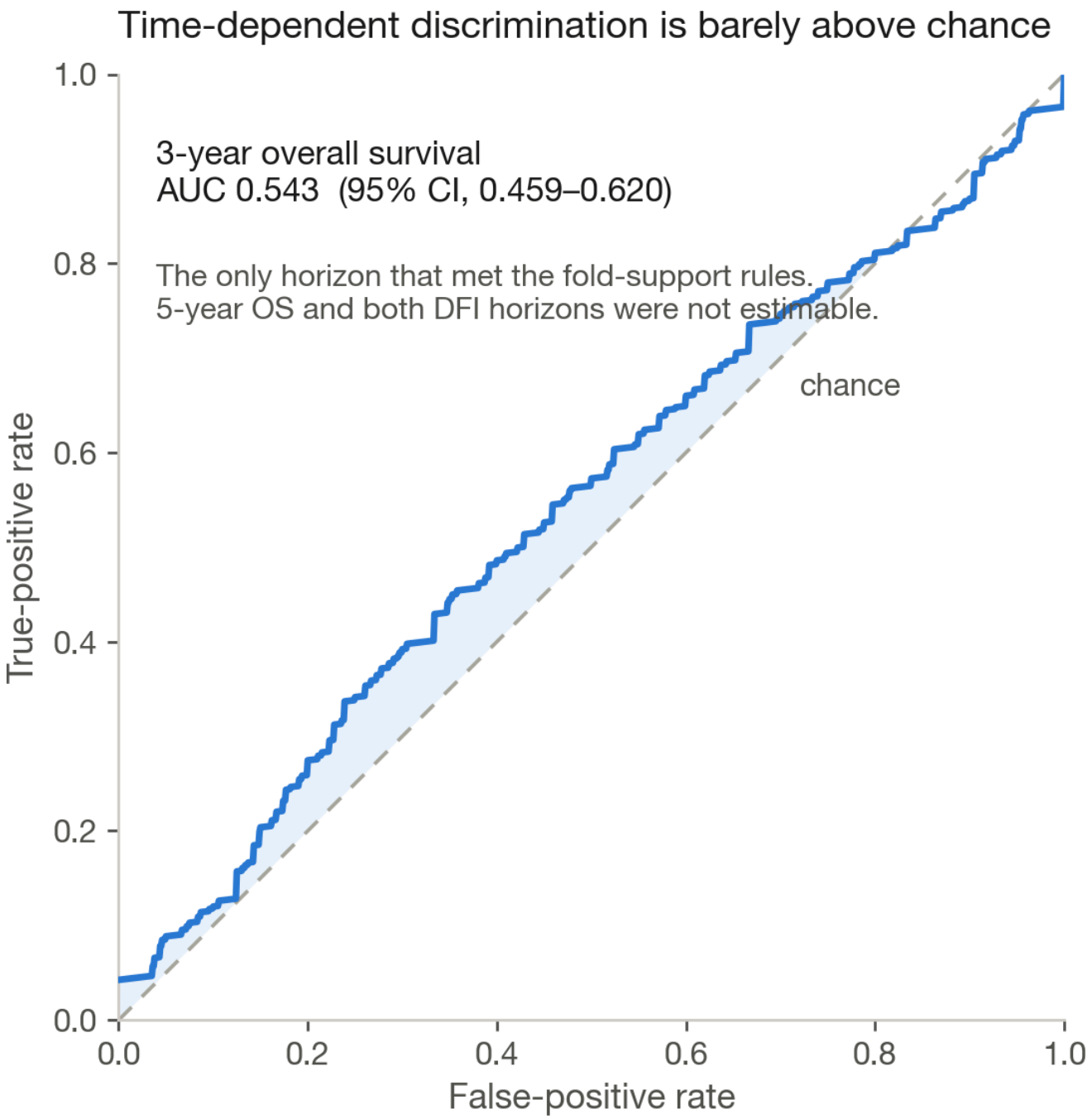


Fig. 5 | Cumulative/dynamic ROC for the panel in all-stage overall survival at three years, the only horizon meeting the fold-support rules, with AUC 0.543 (95% CI, 0.459–0.620). Fold curves were averaged using case–control pair weights within seeds, then across seeds. An internal cross-validation result, not a validation.

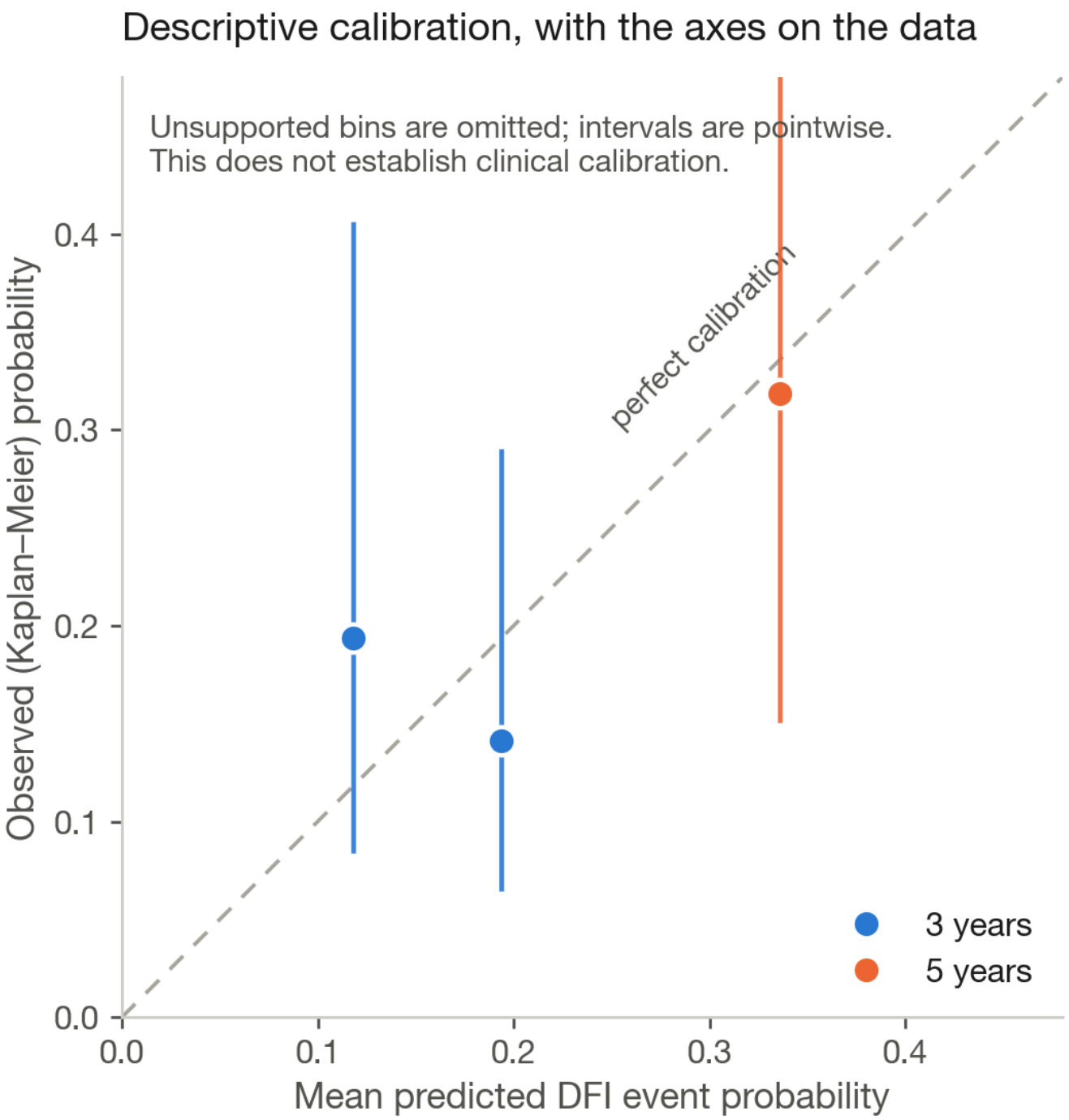


Fig. 6 | Descriptive primary-DFI calibration of the panel. Predicted event probabilities were averaged across held-out repetitions and binned; observed probabilities are Kaplan–Meier estimates with pointwise 95% intervals, and unsupported bins are omitted. These bin criteria differ from the AUC support rules, and the display does not establish reliable clinical calibration.

9. The median split

The primary-cohort expression median was 1.14059 on the $\log_2$(CPM+1) scale. The low-expression group held 69 patients and 12 DFI events; the high-expression group held 68

patients and 6 events. The unadjusted log-rank comparison was not significant (p = 0.115), and the curves cross during sparse late follow-up.

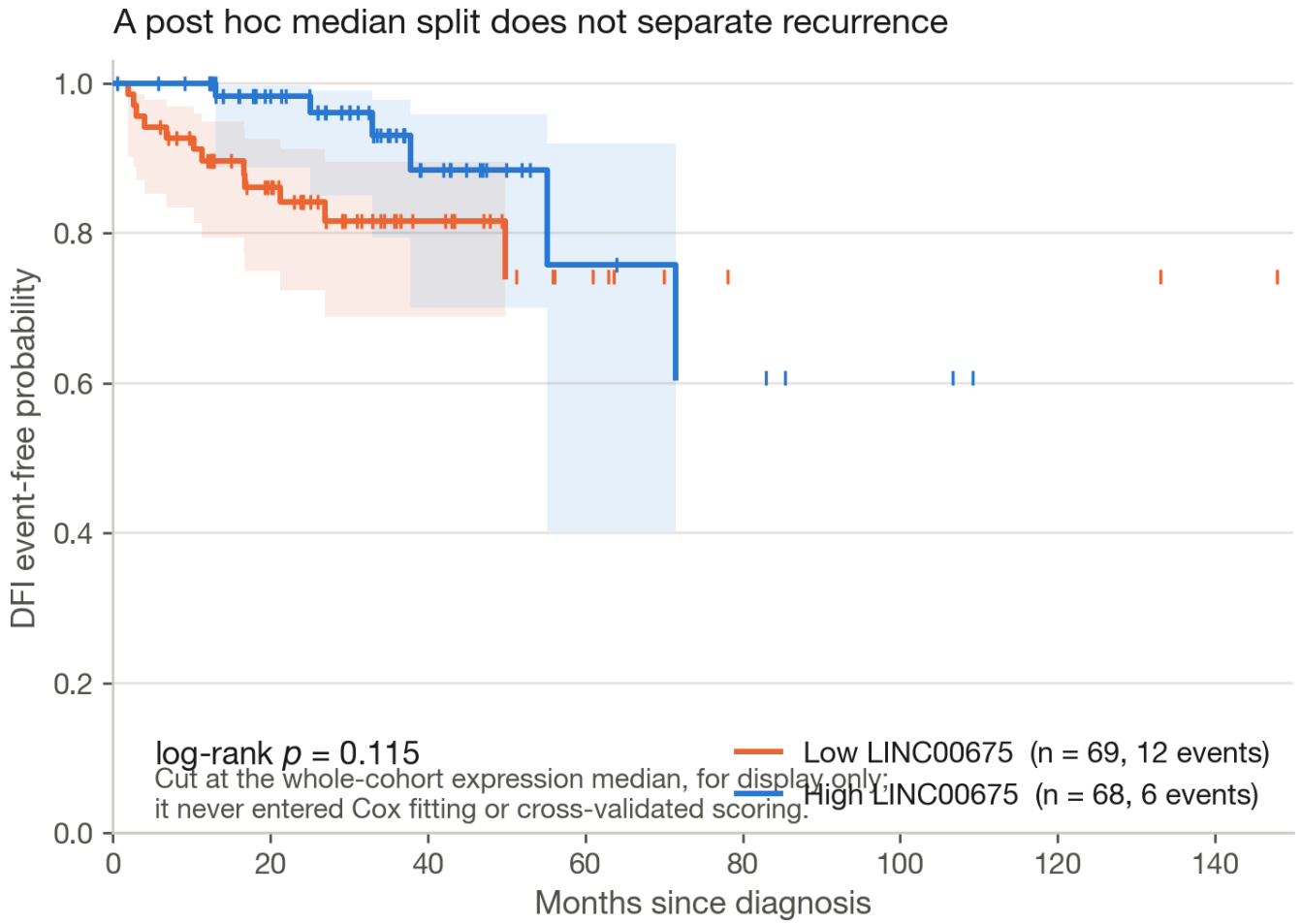


Fig. 7 | Post hoc LINC00675 median split in the primary cohort: 69 patients and 12 events at or below the median, 68 patients and 6 events above it; log-rank p = 0.115. The cutoff was computed on the whole cohort for display only and never entered Cox fitting or cross-validated scoring. The curves cross during sparse late follow-up.

This cutoff was computed on the whole cohort for display only and never entered Cox fitting or cross-validated scoring. A dichotomised whole-cohort comparison and continuous-expression held-out discrimination answer different questions, and failing the first does not invalidate the second. But a reader looking for confirmation that LINC00675 separates recurrence will not find it here either.

10. The comparison the study was missing

The decisive comparison was absent from the submitted manuscript, and it is the one this version adds.

The panel had a null: 1,000 random four-gene panels through the same pipeline. LINC00675 had none. There was therefore no calibrated answer to the only question that matters about 0.700 — how often does a single gene drawn at random, fitted the same way on the same 137 patients and the same 25 folds, discriminate this well? The methods of §4.1 describe how that null was built and the checks it had to pass first.

Against all 1,000 random genes, LINC00675 sits at the 99.7th percentile: 3 draws of 1,000 reach 0.700, a one-sided empirical probability of 0.004. That figure overstates the case, and the reason is the same one that distorted the panel comparison. Of the 1,000 single genes, **710 produce no model at all** — they fail the measurement gate, or their fitted coefficient is numerically zero — and score exactly 0.500 by construction. A null that is 71% inert is not a fair opponent.

Restricting to genes that could have been candidates fixes this. Of the 1,000 draws, 279 pass the measurement gate in all 25 outer folds, which is what all four candidate transcripts do. This subset is selected on expression alone, with no reference to outcome. Its mean C-index is 0.510 and its median 0.503, and none of it is degenerate.

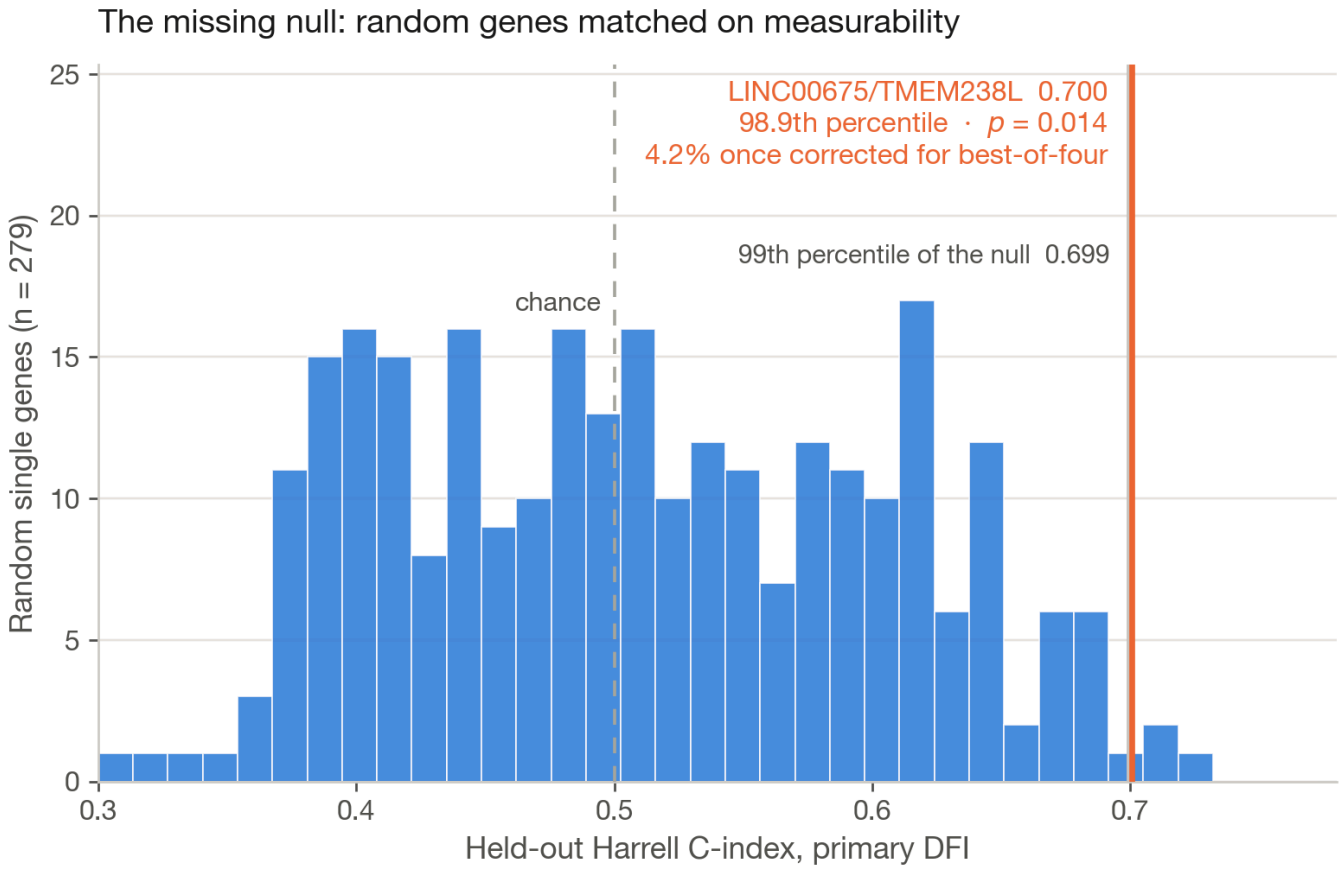


Fig. 8 | The random single-gene null the submitted study lacked, added by the journal. Of 1,000 genes drawn at random and put through the author's own folds, gate, model and scoring, the 279 that pass the measurement gate in all 25 folds — as all four candidate transcripts do — form the comparator shown. LINC00675/TMEM238L sits at the 98.9th percentile, just above the null's 99th percentile of 0.699. Correcting for its having been the best of four candidates, a result this good arises by chance 4.2% of the time. The re-implementation used to build this null first reproduced all four of the author's published single-marker C-indices and all 100 of his per-fold values exactly.

Within that matched null, 3 of 279 genes reach 0.700, a one-sided probability of **0.014**, placing LINC00675 at the 98.9th percentile. The null's own 99th percentile is 0.6993; LINC00675 scores 0.7005. It is, by a small margin, the kind of result you see about once in a hundred random genes.

Accounting for selection is the last step and the one that matters most, because LINC00675 was not drawn at random — it was the best of four candidates. The matching figure is how often the maximum of four genes drawn from the matched null reaches 0.700, which across 200,000 resamples is **4.2%** (4.21% drawing without replacement, 4.19% with). This is a selection-aware empirical calibration rather than a corrected significance level; §4.1 states what it assumes.

So the answer to the question this study could not previously ask is: yes, but barely. A selection-aware probability of 0.042 clears the conventional threshold with almost nothing to spare, on 18 events, in the cohort where the marker was found. It is enough to justify the cost of an external test. It is not enough to call LINC00675 a recurrence biomarker, and it says nothing about whether the marker adds information to stage, mismatch-repair status, age and treatment, which remains the question that decides clinical usefulness.

11. The two checks the referee asked for

A signed report on the working-paper version was filed by Henry Wang, PRISM's editor-in-chief, recommending minor revision. He is barred from refereeing this manuscript under the journal's own conflict policy — he is at the author's school, and he is the author's co-author on PRISM-1-004 — so the paper remains a working paper and his report is not published as a PRISM peer review. Its substance is acted on here regardless, because a request for a missing diagnostic does not become less correct when the person making it is conflicted. Two of the four requests were computable from the material already in hand, and both are reported whichever way they came out.

11.1 Proportional hazards

The paper leans on Cox models throughout and had never tested their central assumption. The crossing curves in Fig. 7 made that worth checking, because a marker whose hazard ratio drifts over follow-up is not well summarised by one coefficient.

A Grambsch–Therneau correlation test on scaled Schoenfeld residuals, with time taken as the rank of the event times, finds no evidence against proportional hazards for any of the four transcripts in the primary cohort: LINC00467 $\chi^2 = 0.49$ ($p = 0.49$), ELFN1-AS1 $\chi^2 = 0.00$ ($p = 1.00$), LINC00675/TMEM238L $\chi^2 = 1.13$ ($p = 0.29$), HOXB-AS3 $\chi^2 = 0.00$ ($p = 0.98$). With 18 events these tests have little power, so this is an absence of evidence rather than a demonstration of adequacy. What it does rule out is the specific worry raised by Fig. 7: the crossing of a dichotomised comparison does not carry over into the continuous model, which is the model the paper actually fits.

11.2 Is LINC00675 a proxy for stage?

This is the question that decides whether the marker could matter clinically, and the answer in this cohort is unusually clean — and unusually hard to lean on.

Adjusting the full-cohort Cox model for stage moves the LINC00675 coefficient almost not at all: −0.5954 alone, −0.5950 with stage in the model, an attenuation of 0.07%. The adjusted hazard ratio is 0.55 per training standard deviation (95% CI, 0.33–0.92). On the author's own 25 folds, held-out discrimination is 0.700 for LINC00675 alone and 0.669 for stage plus LINC00675, against 0.405 for stage alone.

The reason to distrust that comparison is in the last number. **Stage does not predict disease-free interval in this cohort at all.** Its held-out C-index of 0.405 is below chance, and its full-cohort coefficient is a hazard ratio of 0.985 for stage III against stage II — indistinguishable from nothing. Adding a covariate that carries no signal is also why the two-variable model scores *below* the marker alone.

So LINC00675 is not a proxy for stage, and the claim is close to vacuous: in 137 stage II/III patients with 18 events, stage is not a competitor worth beating. The test the referee actually wants — whether the marker survives alongside stage, mismatch-repair status, age and treatment — cannot be run here, both because those covariates were not assembled and because 18 events will not support the model. It belongs in the external cohort, where stage should behave the way the literature says it does.

12. What this study establishes

A panel of four micropeptide-source RNAs, chosen on documented peptide biology rather than by screening, does not predict disease-free interval in TCGA colon adenocarcinoma well enough to pursue. It missed a target set in advance and did not separate from randomly drawn panels under an identical procedure, in the primary cohort and in both secondary ones. That is a clean prespecified negative result, and it is the finding of this paper.

Three further things follow, none of them a biomarker claim. First, mechanistic selection did not concentrate prognostic signal: most of these transcripts discriminated at or below chance. Second, combining four transcripts was worse than using the best one, which identifies a failed strategy without identifying its statistical cause — the final

refit kept only LINC00675, but the held-out predictions came from fold-specific models that often retained others, and feature inclusion, fitted direction and risk ties are all plausible contributors. Third, LINC00675 now has a calibrated standing it did not have before: against genes that could have been candidates, and after correcting for having been the best of four, a C-index of 0.700 arises by chance about 4% of the time.

Two referee-requested checks did not weaken it. Proportional hazards survives testing for all four transcripts, and adjusting for stage moves the LINC00675 coefficient by 0.07%, so the marker is not a restatement of stage. The second of those is worth less than it sounds: stage predicts nothing in this cohort, at a held-out C-index of 0.405, so there was little for the marker to be confounded by.

That is the strongest statement the data support, and it is deliberately a weak one. A selection-aware calibration of 0.042 on 18 events is a reason to run one external test, not a result. What this study still does not establish is that LINC00675 predicts recurrence: it shows no signal for overall survival, does not separate recurrence on a median split, has never been seen outside this cohort, and has never been tested against a clinical model in which stage behaves as the literature says it should. Bulk abundance of this transcript could track epithelial differentiation or tumour cell fraction with no change in peptide abundance at all. RNA is not peptide: no immunohistochemistry or targeted mass spectrometry was performed, and nothing here can establish a FORCP-mediated mechanism.

13. Limitations

Eighteen events in the primary cohort is the limitation from which most others follow: unstable fold estimates, failed horizon support, and wide intervals. Excluding 168 of 305 eligible patients for missing DFI status may also introduce selection bias of unknown direction.

The bootstrap intervals condition on already-fitted cross-validation models and do not capture variability from model fitting, hyperparameter selection or post hoc marker choice. The repeated folds overlap and are not independent replications. Proportional hazards has now been tested and is not contradicted (§11.1), though with 18 events that test has little power.

The single-gene null has limits of its own. It reuses the same 25 folds and the same 137 patients, so it calibrates against sampling variation in this cohort rather than against a new one, and it cannot substitute for external data. Matching on the measurement gate equalises measurability but not expression level, dynamic range or correlation structure, so the matched comparator is fairer than the unmatched one without being an exact like-for-like. The best-of-four calibration treats the four candidates as four draws from that matched population, which is a reference rather than a model of how they were chosen; positive correlation among them would make it conservative and any unrecorded looking would make it optimistic. Age, treatment and mismatch-repair status were never added as covariates, and §11.2 shows why the stage adjustment that was run proves less than it appears to: stage carries no prognostic information in this cohort, so surviving adjustment for it is close to free. Incremental value over a working clinical model remains untested, and for a prognostic marker that is the question that decides clinical usefulness.

No external validation was performed. The expanded DFI cohort overlaps the primary cohort and the OS analysis draws on the same resource, so neither is independent. The random null consists of four-gene panels drawn without expression matching; it contextualises the panel and provides neither a single-marker test nor a substitute for validation.

Gene-level measurement cannot distinguish coding isoforms, translation efficiency or protein turnover, and the historical lncRNA label should not be read as a contemporary coding classification for TMEM238L. DFI is used exactly as supplied by the Clinical Data Resource and is not interchangeable with similarly named endpoints [12].

TCGA's racial imbalance is documented [13]. In these cohorts White patients were a majority among those with recorded race, and race was unavailable for many. Generalisability across populations and care settings is untested, and the study is confined to colon adenocarcinoma; nothing here extends to rectal cancer.

14. What would settle it

The cheap screening step is done. LINC00675 passed it narrowly, which is the outcome that justifies spending more and does not justify believing the result.

A locked external test of LINC00675 alone is next. GSE39582 is a practical candidate only if its array probes measure TMEM238L adequately and uniquely; probe mapping, preprocessing, outcome definition and the continuous-marker model must all be fixed in writing before any outcome is examined. Given a selection-aware probability of 0.042 in the discovery cohort, a failed external test should be treated as the more likely outcome and reported either way.

The decisive result is not the external association on its own. It is whether LINC00675 adds reproducible information beyond stage, mismatch-repair status, age and treatment. §11.2 shows it is not a proxy for stage *here*, but in a cohort where stage itself carries no prognostic information that is a low bar, and the test has to be repeated where the clinical model actually works. Reproducing an unadjusted association would leave the clinical question exactly where it is now.

One cheaper check is also worth doing first, because it could remove the marker without any new cohort: whether bulk LINC00675 abundance is explained by tumour purity or by an epithelial-differentiation score computed on the same specimens. If it is, the association has an explanation that has nothing to do with FORCP.

Declarations

Author. Leo Lee is the sole author. He designed the study, specified the candidate panel and endpoints, ran and audited the analysis pipeline, and approves this version.

Funding. None.

Competing interests. None declared by the author.

Ethics. Not applicable. The study analyses de-identified public data from TCGA and the TCGA Clinical Data Resource. No human participants were recruited and no new specimens were collected.

Use of artificial intelligence. This paper was prepared with substantial AI assistance at every stage, and the division of labour is set out here rather than summarised.

For the underlying study, as disclosed by the author: OpenAI ChatGPT (GPT-5) assisted with literature retrieval and with identifying candidate micropeptide-source RNAs; OpenAI Codex assisted with writing and debugging the Python pipeline, including CPM

normalisation, nested cross-validated elastic-net Cox regression, the random-panel baseline, bootstrap intervals and figure export. The author states that all code was executed and audited by him and that all cited references were verified against the original publications. The archived package records these as author-supplied attestations that its own automated audit did not independently verify, and they are published here on that basis.

For this version: the manuscript submitted to PRISM carried an AI-assisted Abstract, Discussion and Conclusion together with a declaration that the author would independently rewrite those sections before submission. That rewrite had not been performed — the archived validation record states `human_rewrite_completed: false` — and it has still not been performed. Instead, Claude (Anthropic), acting for the journal, reframed the paper for publication: the title, abstract, section structure, the emphasis in §§5–13 and all prose in those sections are AI-written. No AI system is an author.

None of the author's results were altered. No model of his was refitted and no reported number of his was changed; the Methods describe the same pipeline as the submitted version.

Four quantitative additions are new to this version and were made by the journal. Three are arithmetic on the author's frozen outputs: the count of degenerate random panels and the restricted probability in §5.1, the tally of below-chance estimates in §7, and the independent reproduction of his published empirical p of 0.066933 from `random_panel_scores.tsv` as a check that his frozen outputs and his reported numbers agree.

The fourth is new computation: the random single-gene null of §10 and §4.1, with Fig. 8. This required re-downloading the expression matrix, whose SHA-256 was checked against the hash in the author's `results.json`, and re-implementing the per-fold Cox fit and concordance count because `lifelines` and `scikit-survival` could not be installed. That re-implementation was gated on reproducing the author's recorded results: all four published single-marker C-indices to the last recorded decimal, all 100 per-fold C-indices, pair counts and coefficients, and the archived per-sample library sizes. It is the journal's computation, not the author's, and the code is in the deposit so that it can be checked or discarded independently of his. The proportional-hazards tests and stage-adjusted models of §11 were added the same way, after peer review asked for them, and their code is deposited beside it.

Peer review. This paper is published as a working paper and has not been peer reviewed. See the note on the article page.

Data and code availability

The study uses the public TCGA-COAD expression matrix from UCSC Xena and the TCGA Clinical Data Resource. Neither is redistributed here. The journal's addition — the single-gene null code, the 1,000 sampled C-indices, the gate mask and the figure script — is in `prism-single-gene-null/` inside the deposit, alongside its validation output. It is separate from the author's material and can be checked or rejected on its own.

The author's reproducibility package is deposited with the journal as coad-micropeptide-package.zip (30 MB). It contains the analysis pipeline, the frozen results, the patient-filtering audits, the cross-validation splits, the random-panel outputs and the figure sources, with SHA-256 hashes for the two required inputs. It is journal-hosted rather than archived, so it carries no DOI and no persistent identifier. The two source matrices are excluded and must be downloaded from the addresses above; the package records their hashes so a rerun can be checked against the analysed snapshot. Proposed validation data are at GEO accession GSE39582.

Figures are renumbered in this version to follow the argument, and all of them were redrawn by the journal in one house style; the author's originals remain in the archive. Figures 5 and 6 plot coordinates parsed directly out of his PGFPlots sources, so they show exactly the numbers he published. Figures 2 and 7 rebuild Kaplan–Meier curves from his per-patient held-out predictions, and the group sizes and event counts were checked against his own figure legends before anything was drawn. Figure 3 replaces a forest plot of a single coefficient with the per-fold retention counts reported in §6, which carry more information in the same space. The correspondence is: Fig. 1 ← fig4_null; Fig. 2 ← fig1_km; Fig. 3 ← fig3_forest, recast; Fig. 4 ← fig6_single; Fig. 5 ← fig2_roc; Fig. 6 ← fig5_calib; Fig. 7 ← fig7_single_km. Fig. 8 has no counterpart: it is the journal's addition. The drawing code is deposited as `prism-figures/figs.py`. The figures themselves are unaltered, recompiled from the author's TikZ/PGFPlots sources.

References

1. Chothani, Sonia P.; Adami, Eleonora; Widjaja, Anissa A.; et al. (2022). A high-resolution map of human RNA translation. *Molecular Cell* 82(15), 2885–2899.e8. DOI: 10.1016/j.molcel.2022.06.023.

2. Shi, Chengyu; Liu, Fangzhou; Su, Xinwan; et al. (2025). Comprehensive discovery and functional characterization of the noncanonical proteome. *Cell Research* 35(3), 186–204. DOI: 10.1038/s41422-024-01059-3.

3. Deutsch, Eric W.; Kok, Leron W.; Mudge, Jonathan M.; et al. (2026). Expanding the human proteome with microproteins and peptideins. *Nature* 654, 813–825. DOI: 10.1038/s41586-026-10459-x.

4. Ge, Qiwei; Jia, Dingjiacheng; Cen, Dong; et al. (2021). Micropeptide ASAP encoded by LINC00467 promotes colorectal cancer progression by directly modulating ATP synthase activity. *Journal of Clinical Investigation* 131(22), e152911. DOI: 10.1172/JCI152911.

5. Zheng, Caishang; Wei, Yanjun; Zhang, Peng; et al. (2023). CRISPR–Cas9-based functional interrogation of unconventional translatome reveals human cancer dependency on cryptic non-canonical open reading frames. *Nature Structural & Molecular Biology* 30(12), 1878–1892. DOI: 10.1038/s41594-023-01117-1.

6. Li, Xiao Ling; Pongor, Lőrinc; Tang, Wei; et al. (2020). A small protein encoded by a putative lncRNA regulates apoptosis and tumorigenicity in human colorectal cancer cells. *eLife* 9, e53734. DOI: 10.7554/eLife.53734.

7. Huang, Jin-Zhou; Chen, Min; Chen, De; et al. (2017). A peptide encoded by a putative lncRNA HOXB-AS3 suppresses colon cancer growth. *Molecular Cell* 68(1), 171–184.e6. DOI: 10.1016/j.molcel.2017.09.015.

8. Zhu, Song; Wang, Ji-Zhong; Chen, De; et al. (2020). An oncopeptide regulates m⁶A recognition by the m⁶A reader IGF2BP1 and tumorigenesis. *Nature Communications* 11, 1685. DOI: 10.1038/s41467-020-15403-9.

9. Marisa, Laetitia; de Reyniès, Aurélien; Duval, Alex; et al. (2013). Gene expression classification of colon cancer into molecular subtypes: characterization, validation, and prognostic value. *PLoS Medicine* 10(5), e1001453. DOI: 10.1371/journal.pmed.1001453.

10. Gray, Richard G.; Quirke, Philip; Handley, Kelly; et al. (2011). Validation study of a quantitative multigene reverse transcriptase-polymerase chain reaction assay for assessment of recurrence risk in patients with stage II colon cancer. *Journal of Clinical Oncology* 29(35), 4611–4619. DOI: 10.1200/JCO.2010.32.8732.

11. Goldman, Mary J.; Craft, Brian; Hastie, Mim; et al. (2020). Visualizing and interpreting cancer genomics data via the Xena platform. *Nature Biotechnology* 38(6), 675–678. DOI: 10.1038/s41587-020-0546-8.

12. Liu, Jianfang; Lichtenberg, Tara; Hoadley, Katherine A.; et al. (2018). An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell* 173(2), 400–416.e11. DOI: 10.1016/j.cell.2018.02.052.

13. Spratt, Daniel E.; Chan, Tiffany; Waldron, Levi; et al. (2016). Racial/ethnic disparities in genomic sequencing. *JAMA Oncology* 2(8), 1070–1074. DOI: 10.1001/jamaoncol.2016.1854.